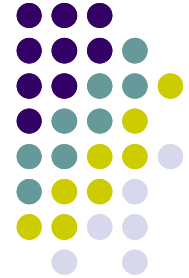


Object Pool

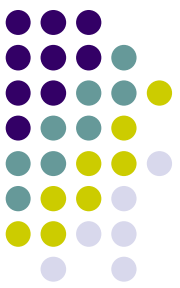


Architecture of Computer Games
SE 456

Ed Keenan

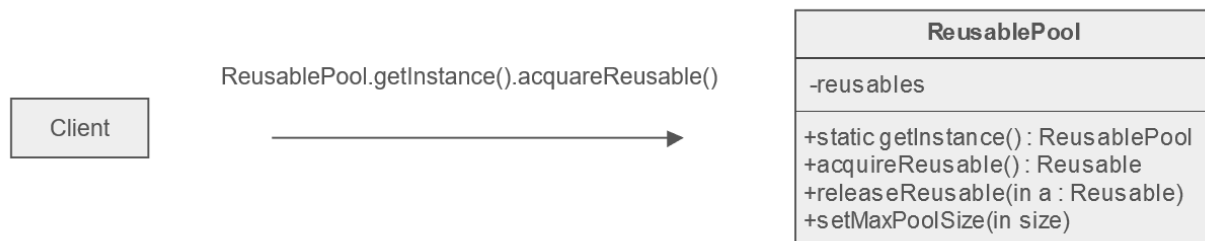
12 January 2022

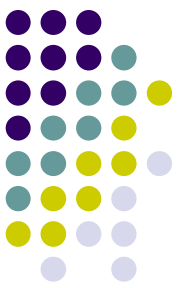
13.0.9.3.9 Mayan Long Count



Object Pooling

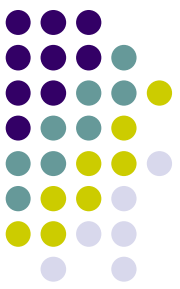
- Intent
 - Reuse and share objects that are expensive to create.
- Remark
 - Best use: cost of initializing a class instance is high, the rate of instantiation of a class is high, and the number of instantiations is finite





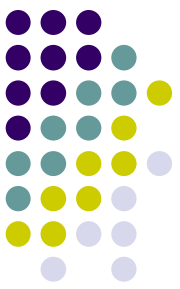
Performance

- Performance issues
 - Cost of the object creation(class instantiation) is a high.
 - Object Pool pattern offer a mechanism to reuse objects that are expensive to create.
- Why to use it?
 - Use an object pool whenever there are clients who needs the same stateless resource which is expensive to create.



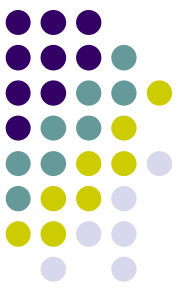
Acquisition

- Client requests an object
 - Pool search for an available object
 - if it was found it will be returned to the client
 - If no object was found then it tries to create a new one.
 - Return the new object



Releasing

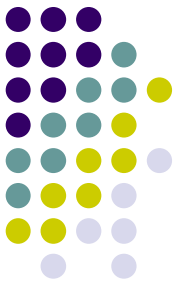
- Client is responsible to release Object back to pool
 - Critical... otherwise its not a reusable pool
- Clients are not aware that it's a recycled object.
 - Clients are not aware that they are sharing the Reusable object.
 - Owners of a new object which comes from the Resource pool
 - Same way from a factory (creational design pattern)
 - Client does not own the object... only borrowing it
 - So it doesn't delete when finish it returns it to the pool.



Implementation

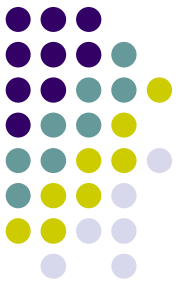
- Limited number of resources in the pool
 - Pool is responsible for sharing and reusing the resources.
 - The way resources are managed affect performance
- Handling situations when creating a new resource fails
 - Need a clear way to communicate that the new object wasn't able to be created

Object Pools



- OODesign.com
 - Object Pool

Thank You!



- Questions?

